

Engineering Mathematics Study Notes: Geometric Series Solved Problems

1. Concept Identification

The latest lecture boards apply the **Geometric Series Test** to multi-term algebraic fractions and numerical fractional series. It introduces two core algebraic strategies for beginners:

- **Splitting the Numerator:** Distributing a shared denominator across a sum of terms to break one complex series into two separate geometric series.
 - **Base Exponential Form Conversion:** Recognizing powers of numbers (like 4, 16, 64) to convert a raw list of fractions into standard geometric index form (r^n).
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2. Mathematical Representations

Linearity Property of Summation

An infinite series containing added terms can be mechanically separated into independent infinite series:

$$\sum_{n=1}^{\infty} (u_n + v_n) = \sum_{n=1}^{\infty} u_n + \sum_{n=1}^{\infty} v_n$$

Index Laws for Powers

Fractions with shared exponents can be consolidated into a single base ratio:

$$\frac{a^n}{b^n} = \left(\frac{a}{b}\right)^n$$

3. The Algorithm (Procedural Recipe)

To solve multi-term or implicit exponential series, follow this exact structural procedure:

Step 1: Simplify into Independent Fractional Terms

If the numerator has multiple added components over a single denominator, split the fraction completely.

Step 2: Extract Common Exponents

Group the individual numerator and denominator powers into a single fraction raised to the power of n . Simplify the internal fraction to its lowest terms.

Step 3: Separate into Independent Geometric Series

Distribute the summation symbol ($\sum$) to each individual exponential term.

Step 4: Extract Parameters (a and r) for Each Sub-Series

For each independent series, identify the first term (a) by substituting the starting index (e.g., $n = 1$) and find the common ratio (r) by inspecting the base of the exponent.

Step 5: Run Individual Convergence Evaluations

Test the absolute value of each common ratio ($|r|$) against the threshold value of 1.

Step 6: Conclude Overall Behavior

Combine the results. If all individual sub-series converge, the entire combined series is **Convergent**. If any single part diverges, the entire expression is **Divergent**.

4. Operational Mechanics & Solutions to Board Problems

Solution to Problem 1: Multi-Term Exponential Fraction

The lecture board presents the following problem:

Test for convergence of the series: $\sum_{n=1}^{\infty} \frac{4^n + 5^n}{6^n}$

Step 1: Split the shared denominator across the numerator terms

$$\sum_{n=1}^{\infty} \left(\frac{4^n}{6^n} + \frac{5^n}{6^n} \right)$$

Step 2: Consolidate the exponents using power laws

$$\sum_{n=1}^{\infty} \left[\left(\frac{4}{6} \right)^n + \left(\frac{5}{6} \right)^n \right]$$

Reduce the fractions to their simplest forms by dividing by their greatest common factors:

$$\sum_{n=1}^{\infty} \left[\left(\frac{2}{3} \right)^n + \left(\frac{5}{6} \right)^n \right]$$

Step 3: Separate into two independent summation loops

$$\sum_{n=1}^{\infty} \left(\frac{2}{3} \right)^n + \sum_{n=1}^{\infty} \left(\frac{5}{6} \right)^n$$

Step 4: Analyze Part 1 ($\sum_{n=1}^{\infty} \left(\frac{2}{3} \right)^n$)

Find the first term by plugging in the starting index $n = 1$:

$$a_1 = \left(\frac{2}{3} \right)^1 = \frac{2}{3}$$

Identify the base ratio under the exponent:

$$r_1 = \frac{2}{3}$$

Evaluate the convergence condition:

$$\left| \frac{2}{3} \right| < 1 \implies \text{Part 1 is Convergent}$$

Step 5: Analyze Part 2 ($\sum_{n=1}^{\infty} \left(\frac{5}{6} \right)^n$)

Find the first term by plugging in the starting index $n = 1$:

$$a_2 = \left(\frac{5}{6} \right)^1 = \frac{5}{6}$$

Identify the base ratio under the exponent:

$$r_2 = \frac{5}{6}$$

Evaluate the convergence condition:

$$\left| \frac{5}{6} \right| < 1 \implies \text{Part 2 is Convergent}$$

Step 6: Final Conclusion

Since both individual sub-series are completely convergent, their sum is stable.

Conclusion: The series is convergent.

Solution to Problem 2: Implicit Numerical Fractional Series

The lecture board presents the following numerical sequence:

$$\text{Test for convergence of the series: } 1 + \frac{3}{4} + \frac{9}{16} + \frac{27}{64} + \dots \infty$$

Step 1: Convert values into base exponent notation

Examine the denominators: 1, 4, 16, 64 are all perfect powers of 4 ($4^0, 4^1, 4^2, 4^3$).

Examine the numerators: 1, 3, 9, 27 are all perfect powers of 3 ($3^0, 3^1, 3^2, 3^3$).

Rewrite the list matching the exponential powers explicitly:

$$1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots \infty$$

This structure matches the standard definition of a geometric series.

Step 2: Extract the parameters directly

Identify the very first number listed in the series:

$$a = 1$$

Identify the constant factor being multiplied to progress from term to term:

$$r = \frac{3}{4}$$

Step 3: Test the stability threshold

Take the absolute value of the ratio parameter:

$$|r| = \left| \frac{3}{4} \right| = \frac{3}{4}$$

Compare this fraction directly against 1:

$$\frac{3}{4} < 1 \implies \text{The absolute value of common ratio is less than 1}$$

Step 4: Final Conclusion

Because the common ratio falls strictly within the convergence window, the infinite addition is guaranteed to settle on a single finite number.

Conclusion: The series is convergent.